

E-COMMERCE DATABASE SYSTEM

Requirement Analysis

1. Project Overview

This project is a simple E-Commerce Database System. It stores details about customers, products, sellers, orders and payments. It helps to manage online shopping information easily.

2. E-Commerce Business Process

- Customer registers in the shopping application.
- Customer views products.
- Customer selects a product and places an order.
- Payment is made for the order.
- Seller receives the order and processes it.
- The product is delivered to the customer.

3. Customer Management Requirements

- Add customer details.
- View customer details.
- Update customer details.
- Delete customer details.
- Store customer orders and payments.

4. Major Entities

The five main entities are:

- Customer
- Product
- Seller
- Order
- Payment

5. Entities and Attributes

Entity	Attributes
Customer	Customer_ID, Name, Email, Phone, Address
Seller	Seller_ID, Seller_Name, Email, Phone
Product	Product_ID, Product_Name, Price, Stock, Seller_ID
Order	Order_ID, Customer_ID, Product_ID, Quantity, Order_Date, Status

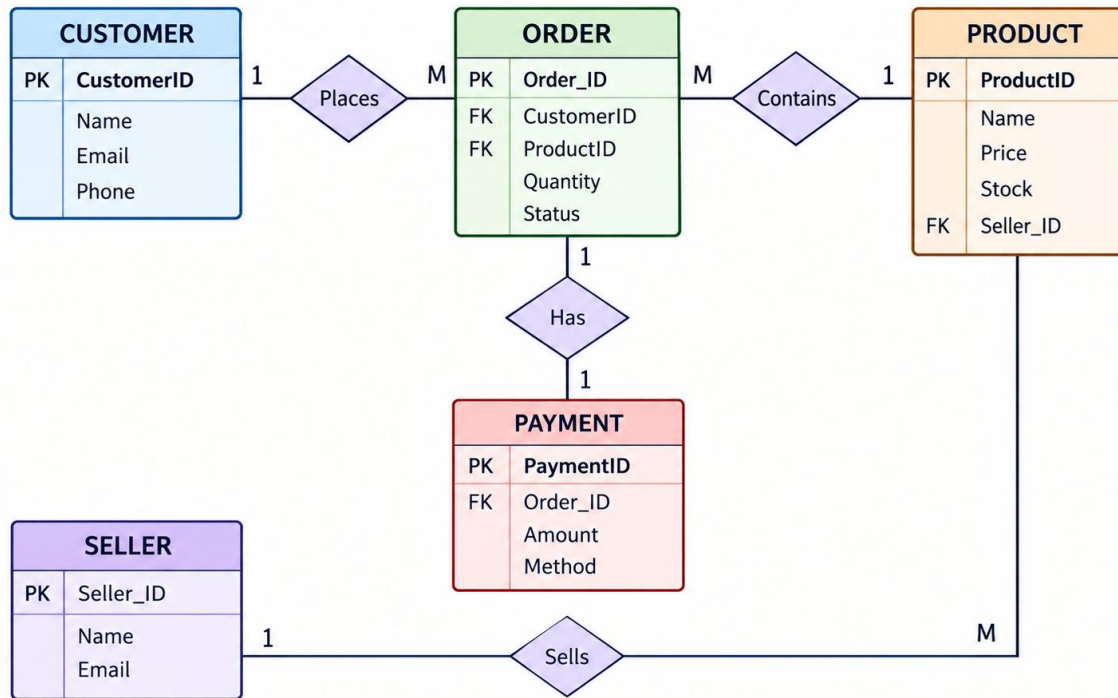
Payment	Payment_ID, Order_ID, Payment_Date, Amount, Payment_Method, Payment_Status
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6. Relationships Between Entities

- Customer → Order: One customer can place many orders.
- Seller → Product: One seller can sell many products.
- Product → Order: A product can be ordered by customers.
- Order → Payment: An order can have a payment.

7. ER Diagram

ER Diagram - E-Commerce Database



8. Functional Requirements

- Customer registration and management.
- Product management.
- Seller management.
- Order management.
- Payment management.
- Search and display records.
- Update and delete records.
- Generate simple database views.

9. Review Questions and Answers

Q. What is a primary key?

Answer: It uniquely identifies each record in a table.

Q. What is a foreign key?

Answer: It connects two tables.

Q. What is NOT NULL?

Answer: It makes a column compulsory.

Q. What is UNIQUE?

Answer: It prevents duplicate values.

Q. What is CRUD?

Answer: Create, Read, Update and Delete.

Q. What is a view?

Answer: A view is a virtual table based on an SQL query.

10. SQL Script

10.1 Create Database

```
CREATE DATABASE ecommerce;  
USE ecommerce;
```

10.2 Create Tables

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Name VARCHAR(50) NOT NULL,  
    Email VARCHAR(100) NOT NULL UNIQUE,  
    Phone VARCHAR(15) UNIQUE,  
    Address VARCHAR(100) NOT NULL  
);  
  
CREATE TABLE Seller (  
    Seller_ID INT PRIMARY KEY,  
    Seller_Name VARCHAR(50) NOT NULL,  
    Email VARCHAR(100) NOT NULL UNIQUE,  
    Phone VARCHAR(15) UNIQUE  
);  
  
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR(100) NOT NULL,  
    Price DECIMAL(10,2) NOT NULL,  
    Stock INT NOT NULL,  
    Seller_ID INT,  
    FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)
```

```

);

CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT,
    Product_ID INT,
    Quantity INT NOT NULL,
    Order_Date DATE NOT NULL,
    Status VARCHAR(20) NOT NULL,
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
    FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
);

CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT,
    Payment_Date DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    Payment_Method VARCHAR(30) NOT NULL,
    Payment_Status VARCHAR(20) NOT NULL,
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)
);

```

10.3 Insert Sample Data

```

INSERT INTO Customer VALUES
(1,'Arun','arun@gmail.com','9876543210','Chennai'),
(2,'Priya','priya@gmail.com','9876543211','Madurai'),
(3,'Rahul','rahul@gmail.com','9876543212','Coimbatore'),
(4,'Divya','divya@gmail.com','9876543213','Trichy'),
(5,'Kavin','kavin@gmail.com','9876543214','Salem'),
(6,'Meena','meena@gmail.com','9876543215','Chennai'),
(7,'Vijay','vijay@gmail.com','9876543216','Erode'),
(8,'Anu','anu@gmail.com','9876543217','Madurai'),
(9,'Siva','siva@gmail.com','9876543218','Tirunelveli'),
(10,'Nila','nila@gmail.com','9876543219','Thanjavur');

```

```

INSERT INTO Seller VALUES
(1,'Tech Store','tech@gmail.com','9000000001'),
(2,'Fashion Hub','fashion@gmail.com','9000000002'),
(3,'Home Store','home@gmail.com','9000000003'),
(4,'Mobile World','mobile@gmail.com','9000000004'),
(5,'Book House','book@gmail.com','9000000005');

```

```

INSERT INTO Product VALUES
(1,'Laptop',55000,10,1),
(2,'Smartphone',25000,20,4),
(3,'Headphones',2000,30,1),
(4,'T-Shirt',800,50,2),
(5,'Shoes',1500,25,2),
(6,'Table Lamp',1200,15,3),
(7,'Keyboard',1500,20,1),
(8,'Notebook',100,100,5),
(9,'Smart Watch',4000,15,4),
(10,'Backpack',1800,20,2);

```

10.4 SELECT Queries

```

SELECT * FROM Customer;

```

```
SELECT * FROM Product;
```

```
SELECT * FROM Product  
WHERE Price > 2000;
```

```
SELECT * FROM Orders  
WHERE Status = 'Delivered';
```

10.5 UPDATE Queries

```
UPDATE Product  
SET Price = 57000  
WHERE Product_ID = 1;
```

```
UPDATE Orders  
SET Status = 'Delivered'  
WHERE Order_ID = 4;
```

10.6 DELETE Queries

```
DELETE FROM Product  
WHERE Product_ID = 10;
```

10.7 Views

```
CREATE VIEW Customer_Order_View AS  
SELECT Customer.Name, Orders.Order_ID, Orders.Status  
FROM Customer  
JOIN Orders  
ON Customer.Customer_ID = Orders.Customer_ID;
```

```
SELECT * FROM Customer_Order_View;
```

11. Sample Program and Output

```
SELECT c.Name, p.Product_Name, o.Quantity, o.Status  
FROM Customer c  
JOIN Orders o ON c.Customer_ID = o.Customer_ID  
JOIN Product p ON o.Product_ID = p.Product_ID;
```

Name	Product	Quantity	Status
Arun	Laptop	1	Delivered
Priya	Smartphone	1	Shipped
Rahul	Headphones	2	Delivered

12. Output Screenshots

After running the SQL commands, insert screenshots of:

- Database creation
- Table creation
- Inserted data
- SELECT output
- UPDATE output
- DELETE output
- View output